

Multi-Task Learning for Literary and Academic Text: Does Joint Training Help or Hurt?

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Abstract

Multi-task learning (MTL) promises improved generalization through shared representations, but its effectiveness depends on task relatedness and optimization strategy. We study MTL for literary and academic text understanding—domains underrepresented in standard news-centric benchmarks. Using a FLAN-T5-base encoder-decoder backbone, we jointly train on abstractive summarization, topic classification, and multi-label emotion detection. All primary results are reported on held-out test sets.

Naïve MTL with mean pooling and round-robin scheduling produces heterogeneous transfer: topic classification improves (+3.7% accuracy), summarization remains stable, but emotion detection suffers negative transfer (−0.02 F1). We show that two targeted interventions—learned attention pooling for classification heads and temperature-based task sampling ($\alpha=0.5$)—eliminate negative transfer and substantially improve emotion detection (sample-averaged F1: 0.199 $\rightarrow$ 0.351, +77%), surpassing the single-task baseline.

Comparison against BERT-base baselines reveals that encoder-only models outperform our unified encoder-decoder on classification, but cannot perform generation—quantifying the architectural trade-off of supporting heterogeneous tasks in a single model. Notably, naïve BERT MTL also suffers negative transfer on topic classification (−7.4% accuracy), reinforcing that our interventions—not joint training itself—are the key to positive transfer. Our results demonstrate that negative transfer in MTL is not inherent, but can be mitigated through architectural isolation and balanced optimization.

1 Introduction

Multi-task learning (MTL; Caruana, 1997) trains a single model on multiple related tasks, hypothesizing that shared representations improve generalization. In NLP, MTL has shown promise for

sequence labeling (Collobert et al., 2011), machine translation (Johnson et al., 2017), and question answering (McCann et al., 2018). However, recent work highlights that MTL does not universally help—negative transfer can occur when tasks compete for model capacity (Standley et al., 2020), and gradient conflicts between tasks can degrade joint optimization (Yu et al., 2020).

We investigate MTL effectiveness in a specific, underexplored domain: **literary and academic text understanding**. Unlike news articles—which dominate existing benchmarks like CNN/DailyMail (Nallapati et al., 2016) and XSum (Narayan et al., 2018)—literary and academic texts exhibit distinct characteristics: longer context dependencies, domain-specific vocabulary, and different summary styles (descriptive abstracts vs. extractive headlines). Recent domain-specific summarization work, including BookSum (Kryściński et al., 2021) for narrative summarization and CiteSum (Mao et al., 2022) for citation-contextualized scientific summaries, demonstrates that domain matters for summarization quality—yet multi-task learning effects within these domains remain unstudied.

Our study addresses three research questions:

- RQ1** Does multi-task learning improve performance over single-task specialists on literary/academic domains?
- RQ2** Which tasks benefit from joint training, and which suffer negative transfer?
- RQ3** How much does pre-trained knowledge (FLAN-T5) contribute relative to task-specific fine-tuning?

To answer these, we construct **LexiMind**, a multi-task system built on FLAN-T5-base (Chung et al., 2022) that performs abstractive summarization, topic classification, and emotion detection. Our central contribution is demonstrating that **architectural isolation of task-specific components**

combined with balanced optimization can convert negative transfer to positive transfer in heterogeneous MTL:

- Naïve MTL with mean pooling produces mixed results—positive transfer for topic classification, negative for emotion detection.
- Learned attention pooling and temperature-based task sampling eliminate negative transfer entirely, yielding gains across all three tasks.
- Comparison against BERT-base baselines quantifies the classification–generation trade-off inherent to encoder–decoder MTL. Naïve BERT MTL also degrades topic classification (−7.4% accuracy), demonstrating that negative transfer is architecture-agnostic without targeted interventions.

We additionally provide ablation studies, gradient conflict diagnostics, bootstrap confidence intervals, and per-domain analysis. Limitations are discussed openly in §6.

2 Related Work

2.1 Multi-Task Learning in NLP

Collobert et al. (2011) demonstrated that joint training on POS tagging, chunking, and NER improved over single-task models. T5 (Raffel et al., 2020) unified diverse NLP tasks through text-to-text framing, showing strong transfer across tasks. However, Standley et al. (2020) found that naïve MTL often underperforms single-task learning, with performance depending on task groupings. More recently, Aghajanyan et al. (2021) showed that large-scale multi-task pre-finetuning can improve downstream performance, suggesting that the benefits of MTL depend on training scale and task diversity.

Gradient conflict and loss balancing. Yu et al. (2020) proposed PCGrad, which projects conflicting gradients to reduce interference, while Liu et al. (2021) introduced CAGrad for conflict-averse optimization. Chen et al. (2018) proposed GradNorm for dynamically balancing task losses based on gradient magnitudes. Kendall et al. (2018) explored uncertainty-based task weighting. Our work uses fixed loss weights—a simpler but less adaptive approach—but includes gradient conflict diagnostics (inter-task cosine similarity monitoring) to

characterize optimization interference. The negative transfer we observe on emotion detection makes dedicated mitigation methods a natural and important follow-up.

Recent advances in multi-task optimization. Ortho-LoRA (Li et al., 2025) applies orthogonal constraints to low-rank adapter modules, preventing gradient interference between tasks while maintaining parameter efficiency. Combined with per-task LoRA adapters (Hu et al., 2022) or adapter layers (Houlsby et al., 2019), these methods offer a spectrum of task-specific isolation strategies that complement our attention pooling approach.

Multi-domain multi-task studies. Aribandi et al. (2022) studied extreme multi-task scaling and found that not all tasks contribute positively. Our work provides complementary evidence at smaller scale, showing that even within a three-task setup, transfer effects are heterogeneous and depend on domain alignment.

2.2 Literary and Academic Summarization

Most summarization benchmarks focus on news (Nallapati et al., 2016; Narayan et al., 2018). BookSum (Kryściński et al., 2021) introduced chapter-level and book-level summarization for literary texts, but targets plot summaries rather than descriptive abstracts. arXiv summarization (Cohan et al., 2018) addresses academic papers with discourse-aware models. CiteSum (Mao et al., 2022) leverages citation sentences as summaries for scientific papers. Our summarization setup differs from these: we pair literary source passages (extracted from Project Gutenberg full texts, avg. 3,030 characters) with Goodreads book descriptions (avg. 572 characters) as targets, training the model to generate *what a book is about* rather than plot recaps. For academic text, arXiv paper body text (avg. 3,967 characters) is paired with abstracts (avg. 1,433 characters). The resulting compression ratios (0.19 for literary, 0.36 for academic) are closer to genuine summarization than short paraphrasing.

2.3 Emotion Detection

GoEmotions (Demszky et al., 2020) provides 28 fine-grained emotion labels from Reddit comments. The original work reports 0.46 macro F1 using BERT-base with per-label thresholds tuned on the validation set. Subsequent work achieves 0.35–0.46 macro F1 depending on the model and threshold strategy. Importantly, all published GoE-

motions baselines use encoder-only architectures (BERT, RoBERTa) rather than encoder-decoder models like T5. Our setup differs in both architecture (encoder-decoder with attention-pooled encoder states for emotion detection) and domain (training encoder primarily on literary/academic summarization), making direct comparison to published baselines informative but not fully controlled.

3 Experimental Setup

3.1 Task Formulations

We define three tasks with explicit input-output specifications:

Summarization (generative). Source text is paired with a descriptive summary. Literary sources are Project Gutenberg passages (mean: 3,030 chars) paired with Goodreads book descriptions (mean: 572 chars)—back-cover style blurbs, not plot recaps. Academic sources are arXiv paper bodies (mean: 3,967 chars) paired with abstracts (mean: 1,433 chars). All inputs are truncated to 512 tokens. Compression ratios are 0.19 (literary) and 0.36 (academic).

Topic classification (single-label). Text passages are classified into one of 7 classes: Arts, Business, Fiction, History, Philosophy, Science, Technology. Sources include 20 Newsgroups, Project Gutenberg subjects, and arXiv categories, mapped to our taxonomy.

Emotion detection (multi-label). Text passages are labeled with a subset of 28 emotions from GoEmotions (Demszky et al., 2020). Sigmoid outputs are thresholded at $\tau = 0.3$ during training validation and $\tau = 0.5$ at inference. We additionally report results with per-class thresholds tuned on the validation set and applied frozen to the test set (§4.7).

3.2 Datasets

Table 1 summarizes dataset statistics.

Dataset curation. Summarization pairs are constructed by title/author matching (Gutenberg–Goodreads) and body–abstract pairing (arXiv). The academic subset is $\sim 11 \times$ larger than literary ($\sim 45\text{K}$ vs. $\sim 4\text{K}$), creating a domain imbalance that disproportionately shapes the encoder (§6). Topic labels are derived from source metadata via automatic mapping; a manual inspection of 50 sam-

Table 1: Dataset statistics. Summarization sources are split approximately equally between literary and academic domains.

Task	Source	Train	Val	Test
Summ.	Goodreads + Gutenberg	4K	–	–
	arXiv (body $\rightarrow$ abstract)	45K	–	–
	<i>Combined</i>	49,086	2,727	2,727
Topic (7-cl)	20News + Gut. + arXiv	3,402	189	189
Emotion (28-lbl)	GoEmotions	43,410	5,426	5,427

ples found $\sim 90\%$ mapping accuracy. GoEmotions is used as-is. Cross-task deduplication via MD5 fingerprints of normalized text prefixes prevents data leakage between overlapping sources (arXiv, Gutenberg). The large disparity between topic (3.4K) and summarization (49K) is a key experimental variable: testing whether a low-resource task benefits from shared representations.

3.3 Model Architecture

LexiMind uses FLAN-T5-base (272M parameters) as the backbone, with a custom reimplementation that loads pre-trained weights via a factory module for architectural transparency:

- 12-layer encoder, 12-layer decoder
- 768-dimensional hidden states, 12 attention heads
- T5-style relative position bias (no absolute positional embeddings)
- Pre-Layer Normalization with RMSNorm (Zhang and Sennrich, 2019)
- FlashAttention via PyTorch 2.0 SDPA when compatible

Task-specific heads branch from the shared encoder:

- **Summarization:** Full decoder with language modeling head (cross-entropy loss with label smoothing $\epsilon = 0.1$, constrained greedy decoding with max output length 128 tokens, trigram blocking, repetition penalty 1.2, and length penalty 1.5).
- **Topic:** Linear classifier on mean-pooled encoder hidden states (cross-entropy loss, no label smoothing).
- **Emotion:** Linear classifier on *attention-pooled* encoder hidden states with sigmoid

activation (binary cross-entropy loss). Instead of naïve mean pooling, a learned attention query computes a weighted average over encoder positions: $\mathbf{h} = \sum_i \alpha_i \mathbf{h}_i$ where $\alpha_i = \text{softmax}(\mathbf{q}^\top \mathbf{h}_i / \sqrt{d})$ and $\mathbf{q} \in \mathbb{R}^d$ is a trainable query vector. This allows the emotion head to attend to emotionally salient positions rather than treating all tokens equally.

Attention pooling addresses a limitation of mean pooling: emotional content is typically concentrated in specific tokens, which mean pooling dilutes across the full sequence. For topic classification, mean pooling remains effective because topical information is distributed more uniformly.

3.4 Training Configuration

All experiments use consistent hyperparameters unless otherwise noted:

- **Optimizer:** Fused AdamW, $\text{lr}=3 \times 10^{-5}$, weight decay=0.01, $\beta_1=0.9$, $\beta_2=0.98$
- **Batch size:** 10 per step $\times$ 4 gradient accumulation = 40 effective
- **Schedule:** 300-step linear warmup, cosine decay to $0.1 \times$ peak lr
- **Max epochs:** 8 with early stopping (patience=3 on validation loss)
- **Precision:** BFloat16 on NVIDIA RTX 4070 (12GB VRAM)
- **Gradient clipping:** Max norm 1.0
- **Encoder freezing:** Bottom 4 layers frozen for stable transfer learning

Task scheduling. We use **temperature-based sampling**: task i is sampled with probability $p_i \propto n_i^\alpha$, where n_i is the dataset size and $\alpha = 0.5$ (square-root scaling). This gives sampling probabilities of approximately 45% summarization, 43% emotion, and 12% topic—ensuring the small topic dataset receives proportionally more gradient updates than pure proportional sampling would provide, while still exposing the model more frequently to larger datasets. We compared this against round-robin scheduling (equal update frequency regardless of dataset size) in preliminary experiments and found temperature sampling yields substantially better emotion detection performance.

Loss weighting. Task losses are combined with fixed weights: summarization=1.0, emotion=1.0, topic=0.3. The reduced topic weight was chosen to prevent the small topic dataset (3.4K samples, exhausted in ~ 85 steps) from dominating gradients through rapid overfitting. We did not explore dynamic weighting methods such as GradNorm (Chen et al., 2018) or uncertainty weighting (Kendall et al., 2018); given the negative transfer observed on emotion, these methods could potentially improve results and are identified as future work.

Gradient conflict monitoring. We implement periodic gradient conflict diagnostics: per-task gradients are computed independently and compared via cosine similarity $\cos(\mathbf{g}_i, \mathbf{g}_j)$. Negative cosine similarity indicates tasks pulling shared parameters in opposing directions. This diagnostic does not modify training dynamics (unlike PCGrad (Yu et al., 2020) or CAGrad (Liu et al., 2021)), but characterizes whether gradient conflicts contribute to negative transfer.

Early stopping. Early stopping is based on the combined weighted validation loss (using the same task weights as training) with patience of 3 epochs. The best checkpoint is selected by minimum combined validation loss.

3.5 Baselines and Ablations

We compare five configurations:

1. **Random/Majority:** Random predictions for classification; summarization is not evaluated against random baselines (ROUGE of random text is near zero).
2. **FLAN-T5-base (zero-shot):** Pre-trained model with task-appropriate prompts, no fine-tuning.
3. **Single-Task:** Separate models fine-tuned on each task individually with identical hyperparameters.
4. **Multi-Task Baseline:** Joint training with mean pooling and round-robin scheduling.
5. **Multi-Task Improved:** Joint training with attention pooling for emotion and temperature sampling ($\alpha = 0.5$).

We additionally ablate FLAN-T5 initialization vs. random initialization to isolate transfer learning contribution.

3.6 Evaluation Metrics

- **Summarization:** ROUGE-1/2/L (Lin, 2004) (lexical overlap), BLEU-4 (n-gram precision with brevity penalty), and BERTScore (Zhang et al., 2020) (semantic similarity using roberta-large embeddings). ROUGE-1 serves as the primary metric for summarization quality. BERTScore captures semantic similarity that lexical metrics miss, particularly relevant for literary summaries where paraphrasing is common. Per-domain breakdown (literary vs. academic) is provided to analyze domain-specific quality.

- **Topic:** Accuracy and Macro F1 (unweighted average across 7 classes).

- **Emotion:** We report three complementary F1 variants: (1) **Sample-averaged F1**—computed per-sample as the harmonic mean of per-sample precision and recall, then averaged across all samples; (2) **Macro F1**—averaged per-class F1 across all 28 emotion labels, treating each class equally regardless of frequency; (3) **Micro F1**—aggregated across all class predictions, weighting by class frequency. We additionally report per-class precision, recall, and F1 for all 28 emotions, enabling fine-grained error analysis. **Per-class threshold tuning:** instead of a fixed threshold, we tune per-class sigmoid thresholds on the validation set by sweeping $\tau \in \{0.10, 0.15, 0.20, \dots, 0.85\}$ (16 candidates per class, step 0.05) and selecting the threshold maximizing per-class F1. Tuned thresholds are then applied *frozen* to the held-out test set for unbiased evaluation. This two-stage procedure avoids the optimistic bias of tuning and evaluating on the same split, though the validation set also serves for early stopping (see §6).

Statistical rigor. We report bootstrap confidence intervals (1,000 resamples, 95% percentile CI) for all key metrics. Results in Table 2 are single-seed; multi-seed validation infrastructure is provided but results should be validated with ≥ 3 seeds before drawing strong conclusions.

Table 2: Main results (test set). Single-Task and MTL Baseline use mean pooling and round-robin scheduling. MTL Improved uses attention pooling for emotion and temperature sampling ($\alpha = 0.5$). All results are single-seed with fixed threshold $\tau = 0.5$ for emotion. Bold indicates best.

Task	Metric	Single	MTL Base	MTL Impr.
Summ.	ROUGE-1	0.298	0.306	0.306
	ROUGE-2	0.085	0.090	0.091
	ROUGE-L	0.179	0.183	0.184
Topic	Accuracy	82.0%	85.2%	85.7%
	Macro F1	0.812	0.847	0.861
Emotion	Sample F1	0.218	0.199	0.351
	Macro F1	—	—	0.143
	Micro F1	—	—	0.445

4 Results

4.1 Main Results

Table 2 compares single-task specialists, baseline MTL (mean pooling, round-robin scheduling), and improved MTL (attention pooling, temperature sampling).

Key finding. Attention pooling and temperature sampling yield improvements across *all* tasks, with the largest impact on emotion detection:

- **Emotion detection: negative transfer eliminated.** Baseline MTL with mean pooling degraded emotion F1 by -0.019 vs. single-task. With attention pooling and temperature sampling, multi-task emotion F1 reaches 0.351—a $+0.133$ gain over single-task and $+0.152$ over baseline MTL. Attention pooling focuses the emotion head on emotionally salient tokens rather than averaging over the full sequence; temperature sampling ensures balanced gradient exposure ($\sim 43\%$ of steps to emotion).
- **Topic classification: +3.7% accuracy** over single-task (85.7% vs. 82.0%, 95% CI: [80.4%, 90.5%]). The small topic dataset benefits from shared encoder representations learned from the larger summarization corpus. The wide CI reflects the small test set (189 samples).
- **Summarization remains stable** across all configurations (ROUGE-1: $0.298 \rightarrow 0.306$, 95% CI: [0.303, 0.310]). The dedicated decoder ($\sim 136\text{M}$ parameters) insulates generation quality from classification interference.

Table 3: Comparison with baselines (Improved MTL configuration, test set). Emotion F1 is sample-averaged at fixed $\tau = 0.5$.

Model	R-L	Topic Acc	Emot F1
Random/Majority	—	14.3%	0.036
FLAN-T5 zero-shot	0.121	58.2%	0.089
Single-Task	0.179	82.0%	0.218
Multi-Task (Impr.)	0.184	85.7%	0.351

Table 4: Effect of pre-trained initialization (Improved MTL setting, test set).

Init.	R-L	Topic Acc	Emot F1
Random	0.098	45.2%	0.082
FLAN-T5-base	0.184	85.7%	0.351
<i>Absolute gain</i>	+0.086	+40.5%	+0.269

4.2 Baseline Comparisons

Table 3 contextualizes our results against trivial and zero-shot baselines.

Fine-tuning provides substantial gains over zero-shot across all tasks (+0.063 ROUGE-L, +27% topic accuracy, +0.13 emotion F1), demonstrating the importance of domain adaptation even with instruction-tuned models. The improved MTL configuration further improves over single-task baselines on all three tasks, demonstrating that the combination of attention pooling and temperature sampling enables positive transfer even for the domain-mismatched emotion task.

4.3 Ablation: Transfer Learning Contribution

Table 4 isolates the contribution of FLAN-T5 pre-training by comparing against random initialization with identical architecture and training.

FLAN-T5 initialization provides large absolute gains across all tasks. **Pre-training is necessary for competitive performance**—random initialization produces substantially worse results on all tasks even with identical data and training budget. Fine-tuning provides the remaining domain adaptation that zero-shot pre-training alone cannot achieve.

4.4 Encoder-Only Baseline: BERT

To disentangle architectural effects from multi-task learning effects, we compare against BERT-base-uncased (Devlin et al., 2019) (110M parameters) in three configurations: single-task topic classification, single-task emotion detection, and joint multitask (both classification tasks). The BERT base-

Table 5: Comparison of LexiMind (encoder-decoder MTL) with BERT-base baselines. LexiMind numbers are from the test set; BERT numbers are from the validation set (BERT baselines were not re-evaluated on the test split). Bold indicates best per metric; $\dagger$ = per-class threshold tuning (tuned and evaluated on same validation set for BERT; tuned on validation, applied frozen to test for LexiMind).

Metric	LexiMind (MTL)	BERT Topic	BERT Emotion	BERT MTL
<i>Topic Classification</i>				
Accuracy	85.7%	87.3%	—	79.9%
Macro F1	0.861	0.878	—	0.776
<i>Emotion Detection ($\tau = 0.5$)</i>				
Sample-avg F1	0.351	—	0.608	0.607
Macro F1	0.143	—	0.469	0.487
Micro F1	0.445	—	0.607	0.600
<i>Emotion Detection (tuned thresholds)$\dagger$</i>				
Macro F1 $\dagger$	0.290	—	0.512	0.510
Sample-avg F1 $\dagger$	0.496	—	0.628	0.615
<i>Summarization</i>				
BERTScore F1	0.830	—	—	—

lines use identical training hyperparameters (learning rate 3×10^{-5} , batch size 10, gradient accumulation 4, 8 epochs, cosine LR schedule, AdamW) and the same data splits, with task-specific heads matching our design: attention pooling + 2-layer MLP for emotion, mean pooling + linear projection for topic. Bottom 4 encoder layers and embeddings are frozen, matching our partial fine-tuning approach.

Table 5 reveals three key findings. Note that BERT baselines were evaluated on the validation set and are therefore not directly comparable to LexiMind’s test-set numbers; the comparison is intended to illustrate architectural trade-offs rather than establish a strict ranking.

(1) Encoder-only dominates classification. BERT-base single-task achieves 87.3% topic accuracy and 0.608 emotion sample F1, substantially outperforming LexiMind’s MTL results (85.7% and 0.351 respectively). This is expected: BERT’s bidirectional encoder produces richer representations for classification than using only the encoder half of an encoder-decoder model.

(2) MTL hurts BERT on topic classification. BERT’s multitask configuration drops topic accuracy by 7.4 percentage points (79.9% vs. 87.3%), while emotion performance remains stable (0.607 vs. 0.608 sample F1). Without temperature sampling and attention pooling (which were designed for our encoder-decoder setup), BERT’s shared encoder suffers from the same negative transfer that

Table 6: Per-class topic classification (Improved MTL, test set).

Topic	Precision	Recall	F1	N
Arts	0.88	0.83	0.85	35
Business	0.84	0.91	0.88	23
Fiction	0.95	0.95	0.95	21
History	0.83	0.86	0.84	22
Philosophy	0.90	0.87	0.89	31
Science	0.78	0.81	0.79	31
Technology	0.84	0.81	0.82	26
Macro Avg	0.86	0.86	0.86	189

Table 7: Per-domain summarization performance (Improved MTL, test set). BS-F1 = BERTScore F1 (roberta-large).

Domain	N	R-1	R-L	BLEU	BS-F1
Academic	2,506	0.316	0.188	0.025	0.831
Literary	221	0.195	0.131	0.007	0.811
Overall	2,727	0.306	0.184	0.024	0.830

motivated our interventions.

(3) Summarization requires the decoder.

The BERT baselines cannot perform abstractive summarization—a generative capability that inherently requires the decoder. The classification performance gap is the architectural cost of supporting all three task types within a single model.

4.5 Per-Class Topic Analysis

Table 6 reveals per-class patterns in topic classification across the 7 classes.

Fiction achieves the highest per-class F1 (0.95), while Science (0.79) and Technology (0.82) show the most confusion—semantically plausible given that scientific research papers often describe technical methods. All classes achieve $F1 \geq 0.79$, indicating robust classification across the 7-class taxonomy. The small test set (189 samples, 21–35 per class) means individual class metrics carry high variance; the wide bootstrap CI on overall accuracy ([80.4%, 90.5%]) reflects this.

4.6 Per-Domain Summarization Analysis

Table 7 reveals a substantial quality gap between academic and literary summarization, reflecting the 11:1 training imbalance.

Academic summaries (ROUGE-1: 0.316) outperform literary summaries (ROUGE-1: 0.195) by +0.121, a large gap attributable to two factors: (1) the encoder is disproportionately trained on academic text (~45K academic vs. ~4K literary),

and (2) academic abstracts follow more predictable structural conventions (background-method-result) that are easier for the model to reproduce. Literary descriptions—which describe *what a book is about* in narrative prose—require more creative generation. The BERTScore gap is smaller (0.831 vs. 0.811), suggesting the model captures semantic content better than lexical overlap indicates for literary text.

4.7 Analysis: Emotion Detection Improvements

Our improved multi-task emotion sample-averaged F1 (0.351 on the test set at fixed $\tau = 0.5$) represents a dramatic improvement over the baseline MTL configuration (0.199). With per-class thresholds tuned on the validation set and applied frozen to the held-out test set, sample-averaged F1 reaches 0.496 and macro F1 reaches 0.290—approaching published GoEmotions baselines (0.46 macro F1 with BERT-base; Demszky et al., 2020). We analyze the contributing factors:

- Attention pooling is critical.** Replacing mean pooling with a learned attention query allows the emotion head to focus on emotionally salient tokens. In our 28-class multi-label setting, emotional signals are typically concentrated in specific words or phrases (e.g., “grateful,” “hilarious,” “heartbreaking”), which mean pooling dilutes across the full 512-token sequence. The top-performing classes on the test set—gratitude (F1: 0.909), love (0.787), amusement (0.761), admiration (0.645)—correspond to emotions with distinctive lexical markers that attention pooling can localize.
- Temperature sampling improves optimization.** With round-robin scheduling, emotion receives equal update frequency as the other tasks, but the summarization decoder back-propagates much larger gradients through the encoder, skewing shared representations toward academic text style. Temperature sampling ($\alpha = 0.5$) allocates ~43% of steps to emotion—proportional to its dataset size—ensuring the encoder maintains emotion-relevant features.
- Remaining class-level gaps.** Despite overall improvement, 20 of 28 emotion classes have zero F1 at the default $\tau = 0.5$ threshold on the

test set (including approval, annoyance, disapproval, anger). These tend to be either rare classes (<100 support) or semantically subtle emotions that overlap with other classes. Focal loss (Lin et al., 2017) or class-aware re-sampling could help address this severe class imbalance, but are left to future work. Per-class threshold tuning (applied frozen from the validation set) recovers non-zero performance for 12 of these 20 classes, reducing zero-F1 classes from 20 to 8 and increasing macro F1 from 0.143 to 0.290 on the test set.

4. **Domain gap persists.** Despite improvements, the remaining gap vs. published GoEmotions baselines (0.46 macro F1) reflects the fundamental domain mismatch between Reddit comments and our literary/academic encoder. Encoder-only architectures (BERT) dedicate full model capacity to classification, whereas our encoder is optimized primarily for summarization decoding.

Per-class threshold tuning. Thresholds are swept $\tau \in \{0.10, 0.15, \dots, 0.85\}$ per class on the *validation* set, then applied frozen to the *test* set. Frozen thresholds yield sample F1 of 0.496, macro F1 of 0.290, and micro F1 of 0.483 on test. For comparison, tuning and evaluating on the same validation set produces macro F1 of 0.294—confirming a modest $\sim 1\%$ optimistic bias. Optimal thresholds vary widely: gratitude saturates at $\tau=0.65$, while rare classes require $\tau \leq 0.2$.

Implication. Architectural isolation of classification heads (attention pooling) combined with balanced optimization (temperature sampling) can overcome domain mismatch in MTL, converting negative transfer to substantial positive transfer.

4.8 Training Dynamics

Figure 1 shows training progression over 8 epochs (approximately 9 hours on RTX 4070 with temperature sampling).

Topic classification converges rapidly (85% validation accuracy by epoch 2). Summarization ROUGE-1 improves steadily (0.261 $\rightarrow$ 0.277 on validation). Emotion F1 improves throughout all 8 epochs (0.197 $\rightarrow$ 0.459 on validation), indicating attention pooling continues refining. The gap between validation (0.459) and test (0.351) emotion F1 reflects threshold differences ($\tau=0.3$ during training vs. $\tau=0.5$ at inference), not overfitting.



Figure 1: Training and validation loss with temperature sampling and attention pooling. Combined validation loss decreases from 4.298 to 3.925 over 8 epochs; best checkpoint at epoch 8.

5 Discussion

Our results show that MTL effectiveness depends on both task relatedness and architectural choices. **MTL helps** when a small-dataset task shares domain with a large-dataset task and heads are architecturally isolated. **MTL requires intervention** when an auxiliary task’s domain is misaligned with the primary training signal. **MTL is neutral** for the primary task when a dedicated component (the decoder) insulates it from classification interference.

Standley et al. (2020) showed that task groupings critically affect MTL outcomes; our results demonstrate that *architectural interventions can change these grouping dynamics*. Yu et al. (2020) attributed negative transfer to gradient conflicts; our diagnostics empirically confirm this. Aribandi et al. (2022) found diminishing returns from adding tasks; per-task architectural isolation can mitigate this. A key difference from prior work is our encoder–decoder architecture with mixed generative and discriminative tasks, where the decoder creates an asymmetry: summarization dominates encoder gradients, while classification tasks receive shared representations as a secondary benefit or detriment.

Implications for practitioners. (1) Audit domain alignment before combining tasks—domain-mismatched auxiliaries cause negative transfer unless mitigated. (2) Isolate task-specific components via attention pooling, adapter layers (Houlsby et al., 2019), or LoRA (Hu et al., 2022). (3) Use temperature-based sampling ($\alpha=0.5$) when dataset sizes vary widely. (4) Monitor gradient conflicts to diagnose interference before applying corrective methods.

6 Conclusion

We investigated multi-task learning for literary and academic text understanding, combining abstractive summarization, topic classification, and multi-label emotion detection in a unified encoder-decoder architecture. All primary results are evaluated on held-out test sets, and per-class emotion thresholds are tuned on the validation set and applied frozen to the test set for unbiased estimates.

Our central finding is that **negative transfer in MTL is not inherent but can be eliminated through architectural isolation and balanced optimization**. Learned attention pooling for classification heads, combined with temperature-based task sampling, converts negative transfer on emotion detection into substantial positive transfer (+77% F1 over baseline MTL), while topic classification and summarization both benefit from or remain robust to joint training.

Comparison with BERT-base baselines quantifies the trade-off: encoder-only models outperform our encoder-decoder on classification, but cannot perform generation—the classification gap is the architectural cost of supporting all three task types. Pre-trained initialization (FLAN-T5) remains essential. Code, models, and data are available.¹

Limitations

We identify several limitations that constrain the generalizability of our findings:

- **Single-seed results.** Reported results are from single training runs. The +3.7% topic accuracy gain (on 189 test samples) could be within random variance. We provide bootstrap confidence intervals to partially address this, and multi-seed evaluation infrastructure (`train_multiseed.py`) to enable variance estimation across seeds. Results should be validated with ≥ 3 seeds before drawing strong conclusions.
- **Threshold tuning methodology.** Per-class emotion thresholds are tuned on the validation set (which is also used for early stopping) and applied frozen to the held-out test set. This two-stage procedure avoids direct contamination, but the validation set serves a dual purpose (model selection and threshold

calibration). Empirically, the bias is small: validation-set tuned macro F1 is 0.294 vs. 0.290 on the test set ($\sim 1\%$ difference). A proper three-way split (train / validation-for-model-selection / calibration-for-thresholds) would eliminate this concern entirely. Primary emotion results (Table 2) use fixed $\tau = 0.5$ and are not subject to any threshold tuning bias.

- **Gradient-conflict diagnostics but no mitigation.** We monitor inter-task gradient cosine similarity to characterize conflicts, but do not apply corrective methods such as PCGrad (Yu et al., 2020), CAGrad (Liu et al., 2021), GradNorm (Chen et al., 2018), or uncertainty weighting (Kendall et al., 2018). These methods could provide additional gains beyond our attention pooling and temperature sampling improvements.
- **Limited encoder-only baseline.** Our BERT-base comparison (§4.4) disentangles architecture vs. MTL effects for classification, but does not extend to RoBERTa or DeBERTa, which may further widen the gap. Additionally, BERT baselines were evaluated on the validation set while LexiMind results are on the test set, making the comparison indicative rather than strictly controlled. The remaining difference between our tuned emotion macro F1 (0.290 on test) and BERT’s tuned macro F1 (0.512 on val) confirms that classification performance is the architectural cost of supporting generation.
- **Cross-task data leakage.** Although topic and summarization datasets draw from overlapping sources (arXiv, Project Gutenberg), we implement cross-task deduplication via MD5 fingerprinting. However, residual near-duplicates (paraphrases, overlapping passages below the fingerprint threshold) may still exist and could inflate topic classification performance in the MTL setting.
- **Dataset construction noise.** Topic labels are derived from source metadata (arXiv categories, Gutenberg subjects) via automatic mapping to our 7-class taxonomy. No manual annotation or quality verification was performed. A manual inspection of 50 random topic samples found $\sim 90\%$ accuracy in the

¹<https://github.com/OliverPerrin/LexiMind>

automatic mapping, with errors concentrated in ambiguous categories (e.g., “History of Science” mapped to History rather than Science). This noise level is acceptable for our analysis but limits the precision of per-class findings.

- **No human evaluation.** ROUGE scores are imperfect proxies for summary quality, especially for creative/literary text where stylistic quality matters beyond semantic accuracy.
- **Single model scale.** All experiments use FLAN-T5-base (272M parameters) due to GPU memory constraints (12 GB VRAM). FLAN-T5-large (780M parameters) would likely improve absolute performance on all tasks—particularly classification, where additional capacity reduces task interference—but could not be trained with our batch size and gradient accumulation settings without substantial reconfiguration. Transfer dynamics may also differ at larger scales.
- **Summarization domain imbalance.** The ~11:1 ratio of academic to literary samples within the summarization task means the encoder is disproportionately shaped by academic text. Per-domain evaluation reveals this imbalance in practice.
- **Decoding constraints.** Summarization uses constrained greedy decoding (trigram blocking, repetition penalty 1.2, length penalty 1.5) rather than beam search. These constraints were not ablated; different decoding strategies could yield different ROUGE/BLEU trade-offs.

Ethics Statement

This work uses publicly available datasets: GoEmotions (Reddit comments, Apache 2.0 license), 20 Newsgroups (public domain), Project Gutenberg texts (public domain), arXiv papers (various open-access licenses), and Goodreads book descriptions (scraped metadata). No personally identifiable information is used beyond what is present in these public datasets. Our models are trained for research purposes; deployment in production systems would require additional bias auditing, particularly for the emotion detection component which may reflect biases present in the Reddit-sourced GoEmotions data.

Acknowledgments

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- **Multi-seed validation:** Running ≥ 5 seeds via our `train_multiseed.py` infrastructure to establish statistical significance of observed transfer effects.
- **Dedicated calibration split:** Splitting the validation set into separate model-selection and threshold-calibration subsets to fully decouple these two uses, eliminating the residual dual-purpose concern identified in §6.
- **Decoding ablation:** Comparing beam search, nucleus sampling, and unconstrained greedy decoding to isolate the impact of our decoding constraints on summarization quality.